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Identifiability-Gated Variance Decomposition of Foundation-Model Performance: A Design and Empirical Feasibility Study

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ABSTRACT Decomposing foundation-model performance variation into family (as a lineage proxy), temporal, and model-specific components requires that these sources be separately identifiable from the observed model population. We formalize the identifiability conditions—crossed family-by-era design, full column rank, bounded condition number, and bounded variance inflation—and develop a pre-measurement gating procedure that detects rank deficiency and collinearity before any variance components are estimated. A simulation study validates a direct restricted-maximum-likelihood estimator, showing it recovers known ground truth to within 2.5 percentage points under balanced occupancy and 5.3 under realistic sparse occupancy, and fails detectably when family and era are nested: three complementary detectors flag aliasing in 100% of repetitions with zero silent coverage. We then apply the gate to an empirical population of 16 real foundation models spanning 5 families and 11 occupied release quarters. The design fails four of five pre-specified gate diagnostics: family-era crossing (two singleton families), full-rank estimability (rank 14 of 15 columns required), numerical conditioning (condition number 4.7×10^{16}), and variance inflation (infinite VIF). The connectedness condition passes. The failure is structural, caused by a singleton family–era confound (Llama is the only model in 2023Q1, making its family indicator identical to the 2023Q1 era indicator). When we fit the variance-component model despite gate failure, the resulting estimates are highly unstable and non-identifiable, with confidence intervals covering the full $[0\%, 100\%]$ range. We further characterize alternative population designs through a systematic design-space analysis, identifying a sufficient design configuration in the evaluated design grid (30 models, 6 families, 8 eras, rank 13/13, $\kappa = 93$, VIF $= 2.1$). The same design-gating principle may be applicable to other crossed random-effects analyses of model populations, and the gating procedure should be applied before any real-data variance-attribution claim.

INDEX TERMS Foundation models, Large language models, Variance decomposition, Identifiability, Mixed-effects models, Model lineage, Experimental design, LLM evaluation

I. INTRODUCTION

WHEN a group of public open-weight language models is given a hard question, they often fail together—missing the same fact, making the same mistake, reaching the same wrong answer. Kim *et al.* [1] document this across more than 350 open and hosted models: when a pair of models both err, they agree on the wrong answer roughly 60% of the time, and the agreement persists across distinct architectures and providers. This correlation has practical consequences: ensembles assume independent error, and deployment pipelines inherit blind spots from related models [28]. Chen [31] shows that pairwise error correlation substantially underestimates the co-failure ceiling—the probability that every member of an ensemble fails a question—by roughly a factor of two across 67 frontier models.

The consequence is that a review pipeline that routes each item to three open-weight models and escalates to a human only when they disagree rests on the assumption that the three fail independently enough that unanimous agreement is informative. If a shared ancestor or a shared training year makes them wrong together on the same items, then unanimity is loudest where it is least trustworthy. The operator cannot detect this from per-model accuracy, which may be excellent for all three.

A. LINEAGE VERSUS ERA

Two explanations compete for the shared failure structure:

- **Lineage:** A model trained on another model's outputs, or on data generated by it, tends to replicate that model's blind spots. Deploying models with independent ancestries reduces the probability that the ensemble fails a question the same way.
- **Era:** Models released at the same time are trained on largely the same scrapes of the web, the same benchmark-cleansed corpora, and the same dominant techniques, so they may pick up the same errors independently of any shared ancestry.

The obstacle to separating them is that lineage and era are temporally entangled: descendants are always released after their ancestors, so release year is simultaneously correlated with lineage and with the training environment. A model released in 2024 shares errors with its siblings by ancestry and with unrelated contemporaries by shared training environment. Observing correlation does not allocate it.

B. THE METHODOLOGICAL PROBLEM

Whether shared failure is mostly inherited or mostly contemporaneous is not an academic question. If shared mistakes trace mostly to lineage, then diversification across model families is a credible mitigation. If they trace mostly to era, then no amount of new ancestry helps, because every model of a given year carries the same errors. The choice between these two conclusions affects how practitioners assemble model portfolios.

The difficulty is that the two explanations are not separable by inspection. Release year lies on the lineage-to-error path: a child model inherits both the errors of its parent and the training environment of its own release date. Lineage and release era are therefore statistically confounded in the observed model population. Any single-number answer that tries to assign shared error "to lineage" or "to era" conflates the two sources.

C. WHY POPULATION DESIGN MATTERS

Variance decomposition is standard in genetics, education, and psychology, but the estimability of variance components depends critically on the design. A nested design—each family confined to one era—makes lineage and era collinear: $\sigma^2_{\text{lineage}}$ and σ^2_{era} cannot be separated from observational data. This is a property of the design, not a data gap: more data within a nested design does not help.

For a crossed random-effects decomposition to be identifiable, the model population must satisfy several structural requirements: the family-era incidence must be crossed (each family appears in multiple eras, and each era contains multiple families), the design matrix must have full column rank, the numerical conditioning must be sufficient for stable estimation, and the variance inflation factors must be bounded. Throughout, we report the condition number of the Gram matrix, $\kappa(X^\top X)$, rather than of X itself; the corresponding condition index on X is therefore $\sqrt{\kappa}$ (Sec. IV). These requirements can be checked before any model is evaluated.

D. RESEARCH GAP

Prior work has extensively studied correlated model behavior and model lineage. Kim *et al.* [1] construct and analyze a large model population, but their objective is correlation measurement—predictors of pairwise error agreement—not variance attribution. PhyloLM [24] reconstructs ancestry from output similarity without decomposing error variance. Total Evaluation Error (TEE) [27] decomposes pipeline measurement facets, not model trait variance. The monoculture literature [22], [23], [25] treats correlated error as a given input to welfare arguments or studies how monoculture claims depend on the choice of model population and independence baseline. Kuai *et al.* [26] measure behavioral entanglement across six families but stop at an independence audit rather than a trait-level variance partition. Recently, the notion of an identifiability audit for LLM evaluation has begun to appear [35], which argues that controlled reasoning-evaluation protocols can be structurally non-identifiable and should be audited before inference. That work audits the identifiability of an *evaluation-protocol/behavioral* estimand; it does not gate the *population design* that governs variance-component estimands. Our contribution targets precisely that distinct object—the structural and numerical identifiability of crossed family-proxy and release-era variance components over a model population—and is, to our knowledge, the first to combine outcome-independent foundation-model population selection with an explicit pre-measurement identifiability gate for that estimand.

The central methodological claim of this study is that variance attribution should be treated as a gated inference problem: the model population must first be shown to support the intended estimand, then the estimator must be validated under comparable designs, and only then should empirical variance components be interpreted.

E. RESEARCH QUESTIONS

RQ1: Under what family-era population structures is the proposed family-proxy-era variance decomposition identifiable?

RQ2: Does the proposed estimator recover known variance components under identifiable and non-identifiable simulated designs?

RQ3: Does the real 16-model empirical population satisfy the identifiability requirements?

RQ4: What population structures provide sufficient conditioning and crossing for future empirical estimation?

F. CONTRIBUTIONS

1) **Formal identifiability framework.** We specify the rank, conditioning, crossing, and connectivity require-

ments that a family-era design must satisfy before variance components can be estimated, separating structural identifiability from numerical stability.

2) **Pre-measurement gating procedure.** We develop a computational test that checks rank, condition number, variance inflation, and design crossing before any model is evaluated, with thresholds validated through simulation.
3) **Simulation validation and failure detection.** We validate the direct REML implementation under identifiable and non-identifiable designs, showing recovery within 2.5pp under balanced and 5.3pp under realistic occupancy, and demonstrate three complementary detectors that flag nested aliasing in 100% of repetitions.
4) **Empirical feasibility and population-design analysis.** We apply the gate to 16 real models, show the gate fails, diagnose the structural deficits, and identify a sufficient design configuration in the evaluated design grid (30 models, 6 families, 8 eras) through systematic design-space analysis.

II. RELATED WORK

A. CORRELATED ERRORS IN LLMS

Kim *et al.* [1] provide large-scale evidence that LLM errors are empirically correlated. Using 349 HuggingFace models with leaderboard-provided question-level MMLU responses and 71 HELM models, they measure pairwise error agreement—the probability that two models produce the same wrong answer conditional on both being wrong. They find mean agreement of 0.423 (HuggingFace) and 0.600 (HELM), with 100% and 97.5% of model pairs exceeding the random-error baseline, respectively. Their pairwise regression on 60,726 HuggingFace pairs identifies shared provider ($\beta = 0.066$), shared architecture ($\beta = 0.076$), model size, and model accuracy as significant predictors of correlated errors. Their population selection is outcome-informed: they begin with the 500 most accurate models on the HuggingFace Open LLM Leaderboard, retain 451 with usable MMLU data, and impose developer/fine-tuner constraints (capping individual uploaders at five models) to prevent any single contributor from dominating, yielding 349 models. Chen [31] extends this line by showing that pairwise error correlation substantially underestimates the co-failure ceiling of an ensemble across 67 frontier models and three combination schemes.

Our work addresses a complementary problem: rather than estimating predictors of pairwise dependence, we ask whether a model population has sufficient crossed family–era structure to support attribution of model-level performance variation to family and release-era components. Kim *et al.*'s outcome-informed selection is appropriate for their correlation survey but would bias a variance-component analysis; our population-selection procedure is explicitly outcome-independent because the target estimand would otherwise be subject to design-stage outcome selection. This distinction motivates an explicit pre-measurement identifiability gate.

B. MODEL LINEAGE AND ANCESTRY

PhyloLM [24] builds phylogenetic trees from output similarity across open and closed models and shows that tree distance predicts benchmark performance. Tracing the Roots [28] reconstructs the evolution of datasets in post-training—83 seed datasets giving rise to 430 datasets and 971 inheritance edges—and shows that benchmark contamination propagates along lineage paths. Kuai *et al.* [26] measure behavioral entanglement across six open-weight families but stop at an independence audit rather than a trait-level variance partition.

C. VARIANCE-COMPONENT AND MIXED-EFFECTS MODELS

The statistical machinery this paper applies is well established. Estimating the variance attributable to each of several crossed grouping factors is the classical variance-components problem, developed for unbalanced designs by Henderson [5] and placed on a likelihood footing by Patterson and Thompson [3] and Harville [2], whose restricted likelihood removes the downward bias that maximum likelihood incurs by ignoring the degrees of freedom spent on fixed effects. Searle, Casella, and McCulloch [6] and Rao and Kleffe [7] give the general theory; Laird and Ware [9] and McCulloch and Searle [8] extend it to the mixed-model and generalized settings. Generalizability theory [4], [14] was built precisely to ask what fraction of observed variation in a measurement is attributable to persons, items, raters, or occasions. We did not find prior work that explicitly poses model lineage and release era as the crossed facets of a model population's error variance.

D. VARIANCE DECOMPOSITION IN EVALUATION PIPELINES

The closest statistical neighbor is the total-evaluation-error line. Messing [27] decomposes the uncertainty of LLM evaluation pipelines—judge-model choice, temperature, and prompt phrasing—into crossed random effects via generalizability theory. The estimator class is the same (crossed random effects, REML), but the grouping factors are properties of the measurement pipeline (prompt, judge, temperature), not of the models themselves. The present work instead treats the model population as the object of study: family/lineage proxy and release era are structural facets of the models, and the target is model-level trait variance rather than pipeline measurement noise. Separately, protocol-level work [35] audits whether a controlled reasoning protocol is identifiable for its own estimand; our gate plays an analogous role for the crossed family-era variance-attribution estimand.

E. MODEL ENSEMBLES AND DIVERSIFICATION

That ensemble benefit depends on error independence rather than on member accuracy is one of the oldest results in machine learning [18]–[20]. Kuncheva and Whitaker [21] evaluate ten diversity measures and find their relationship to ensemble accuracy weaker than the folk theory assumes. This literature almost universally assumes members are trained independently, which is exactly the assumption that fails for a portfolio of public language models.

F. POSITIONING

TABLE I summarizes the gap. The agreement and co-failure work measures that models err together but reports a correlation, not a decomposition. The variance-components and generalizability literature supplies the decomposition but has not, to our knowledge, been applied to a foundation-model population with an identifiability check. The pipeline-variance work decomposes a different object (measurement facets) than ours (model-population structural facets). The lineage-reconstruction work recovers ancestry structure but stops before attributing error variance to it. Protocol-level identifiability audit work [35] is concerned with reasoning-protocol estimands rather than model-population variance attribution. The present contribution differs from all of the above in three respects: (1) our population-selection procedure is explicitly outcome-independent, (2) we apply a pre-measurement identifiability gate that checks structural and numerical requirements before any model is evaluated, and (3) the target estimand is a variance-component decomposition of model-population trait variation rather than a pairwise correlation, a causal lineage effect, or a protocol-level estimand.

III. PROBLEM FORMULATION AND ESTIMANDS

A. POPULATION MODEL

Let the population consist of N open-weight models indexed by $i \in \{1, \ldots, N\}$. Each model has a family $f(i) \in \{1, \ldots, F\}$ and a release quarter $e(i) \in \{1, \ldots, E\}$. Each model receives a continuous per-model trait Y_i measured on a fixed common item set.

The trait model is:

$$Y_i = \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i \tag{1}$$

where μ is the grand mean, $\alpha_f \sim \mathcal{N}(0, \sigma_L^2)$ are independent family (lineage) effects, $\beta_e \sim \mathcal{N}(0, \sigma_E^2)$ are independent era effects, and $u_i \sim \mathcal{N}(0, \sigma_U^2)$ are independent model-specific residuals. The three variance components are $\theta = (\sigma_L^2, \sigma_E^2, \sigma_U^2)$.

In vector form:

$$\mathbf{y} = \mu\mathbf{1} + C\boldsymbol{\gamma} + \mathbf{u}, \quad \boldsymbol{\gamma} \sim \mathcal{N}(\mathbf{0}, \mathrm{diag}(\sigma_L^2\mathbf{1}_F, \sigma_E^2\mathbf{1}_E)) \tag{2}$$

where C is the $N \times (F + E)$ design matrix of family and era indicators. The marginal covariance is:

$$V(\theta) = \sigma_U^2 I + C\ \mathrm{diag}(\sigma_L^2\mathbf{1}_F, \sigma_E^2\mathbf{1}_E)\, C^\top \tag{3}$$

B. OBSERVATIONAL SHARE ESTIMAND

The primary observational estimand is the share vector:

$$\theta_P = \left(\frac{\sigma_L^2}{\sigma_T^2}, \frac{\sigma_E^2}{\sigma_T^2}, \frac{\sigma_U^2}{\sigma_T^2}\right) \tag{4}$$

where $\sigma_T^2 = \sigma_L^2 + \sigma_E^2 + \sigma_U^2$ is the total variance. This is the family (lineage-proxy) share conditional on era grouping—the proportion of performance variance attributable to family membership, holding era as an observed covariate. A causal or mechanistic lineage estimand would require controlled or naturally co-released cohorts and is outside the scope of the present feasibility study.

C. WHAT IS AND IS NOT IDENTIFIED

If the design is rank-deficient, the variance components are aliased: multiple combinations of $(\sigma_L^2, \sigma_E^2, \sigma_U^2)$ produce the same likelihood. A numerical optimizer can converge to a finite parameter vector even when the design is non-identifiable, but the resulting estimates are not uniquely determined by the data. Reporting them as if they were is misleading. The observational estimand θ_P is structurally identified when the design is crossed, connected, and has full column rank. Our operational reporting gate additionally requires bounded condition number ($\kappa \leq 100$) and bounded variance inflation (VIF ≤ 10) based on simulation-calibrated stability criteria; these are necessary for reliable estimation but do not themselves determine mathematical identifiability. Because the measured population contains only 16 model-level observations, the empirical study is designed primarily to evaluate population identifiability rather than to obtain precise variance-component estimates. TABLE II summarizes the estimand and identification strategy.

IV. FORMAL IDENTIFIABILITY CONDITIONS

A. FAMILY-ERA INCIDENCE STRUCTURE

Given a population with F families and E release quarters, the design matrix is $C = [Z_F \mid Z_E]$, where Z_F is the $N \times F$ family indicator and Z_E is the $N \times E$ era indicator. With reference coding (dropping one column from each block), the full model matrix is $X = [\mathbf{1} \mid Z_F^{(-)} \mid Z_E^{(-)}]$, which has $p = 1 + (F - 1) + (E - 1)$ columns.

B. CROSSING

A design is crossed if every family spans at least two release quarters and at least two quarters contain at least two families. Crossing is what separates the two variance components: it ensures that family effects and era effects are not perfectly confounded. A family confined to a single era (as Llama and Qwen are in the measured 16-model population) creates a perfect local confound between that family's effect and its era's effect.

C. CONNECTEDNESS

A design is connected if the bipartite family-era incidence graph—one vertex per family, one per quarter, an edge whenever some model occupies that cell—is connected. Connectedness licenses comparison across the whole population: in a disconnected design the components carry separate, non-comparable grand means, and a partition estimated across them is a partition of an artifact.

D. FULL COLUMN RANK

We use full column rank of the fitted mean-model design matrix, $\mathrm{rank}(X) = p$, as a necessary structural gate for the spec-

TABLE I: Related Work Comparison. Comparison of representative work across six dimensions relevant to family/era variance attribution. Prior work addresses correlated-error measurement, lineage reconstruction, pipeline variance, or behavioral entanglement, but does not combine outcome-independent population selection with pre-measurement identifiability gating for crossed variance-component attribution.

Work	Population/Setting	Dependence Analyzed	Lineage	Temporal Facet	Stats. Attrib.	Pre-analysis Gate
Kim *et al.* (ICML 2025)	349 models (outcome-informed)	Pairwise error agreement	—	—	Pairwise regression	—
Chen (2026)	67 models	Co-failure ceiling	—	—	Pairwise correlation	—
PhyloLM (ICLR 2025)	Open + closed	Output similarity	Phylogeny	—	Tree distance	—
Tracing Roots (ACL 2026)	Post-training	Dataset propagation	Dataset lineage	—	Propagation count	—
Kuai *et al.* (2026)	18 models, 6 families	Behavioral entanglement	—	—	Independence audit	—
Jo *et al.* (2026)	Monoculture	Population dependence	—	—	Pop.-sensitive critique	—
Messing, TEE (2026)	Pipeline facets (prompt, judge, temp.)	Judge/temp. variance	—	—	Crossed REML	—
This work	16 models	Family proxy + era variance	**Family variance**	**Release quarter**	**Crossed REML**	**Yes: rank/κ/VIF**

TABLE II: Estimand and Identification Strategy.

Estimand	Target Quantity	Unit	Identification Condition	Status
θ_P	σ_L^2/σ_T^2	Model	Crossed family-era design, full rank, connected incidence, $\kappa \leq 100$, VIF ≤ 10	**Not identifiable**

ified mean-model parameterization. Rank deficiency means the fixed-effect coefficients are aliased and not uniquely determined. It is important to keep three distinct requirements separate rather than collapsing them into a single "identifiability" condition: (i) *structural mean-model identifiability* is governed by the rank of X; (ii) *variance-component identifiability* additionally requires linear independence of the covariance basis matrices (below); and (iii) *numerical stability* requires bounded conditioning and bounded variance inflation (following subsections). Full rank of X is thus necessary but not sufficient for either (ii) or (iii); each level is checked separately by the gate.

Proposition 1 (Connected crossing implies full incidence rank)**.** *Let $C = [Z_F \mid Z_E]$ be the $N \times (F+E)$ incidence matrix of a connected crossed design. Then C has rank $F+E-1$, the deficiency being exactly the intercept direction $\mathbf{1}$. Connectedness and crossing imply the stated rank property of the incidence design. Variance-component identifiability additionally requires linear independence of the covariance basis matrices $\{I, Z_F Z_F^\top, Z_E Z_E^\top\}$ in the space of $N \times N$ symmetric matrices. We verify this numerically for the 16-model design, a nested reference, and the 30-model sufficient design: in all three cases the covariance-basis rank is 3/3 (full). The binding constraints for the empirical population are therefore design rank and numerical conditioning, not covariance-basis degeneracy.*

E. NUMERICAL CONDITIONING

Even when the design is rank-complete, the condition number quantifies numerical stability. We compute $\kappa(X^\top X)$, the ratio of the largest to smallest eigenvalue of the Gram matrix $X^\top X$. This is the square of Belsley's condition index (which is based on the square-root eigenvalue ratio of X itself). We pre-specify $\kappa_{\max} = 100$ as the threshold for acceptable conditioning. Belsley classifies condition indices above 10 as indicating moderate multicollinearity and above 30 as severe; our threshold corresponds to a condition index of $\sqrt{100} = 10$ on the X scale. This threshold is a design choice validated through simulation, not a universal mathematical constant.

F. VARIANCE INFLATION

The variance inflation factor (VIF) for column j is $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 is the coefficient of determination from regressing column j on all other columns. We require $\max(\text{VIF}) \leq 10$. High VIF indicates that a particular effect is nearly confounded with other effects in the design.

G. WHY RANK ALONE IS INSUFFICIENT

Full rank is a necessary condition for identifiability but not sufficient for stable estimation. A design can have full rank but be so nearly aliased that the components are not recoverable at realistic sample sizes. This is the gap that the simulation stage exists to close: the operational gate combines structural rank (necessary) with numerical conditioning and VIF (stability requirements), and the simulation validates that the thresholds are calibrated to the actual design size.

H. OPERATIONAL GATE

The overall workflow is shown in Fig. 1. The pre-measurement identifiability gate comprises five conditions, divided into structural requirements (which must hold for the design to be theoretically identifiable) and numerical stability requirements (which must hold for reliable estimation):

Structural Gate (S1–S3):

- **S1: Family-era crossing** — Every family appears in more than one era and at least two eras contain more than one family. A family confined to a single era creates a perfect local confound between that family's effect and its era's effect. This is the study's pre-specified operational crossing criterion, chosen to guarantee sufficient within-family and within-era replication for the intended decomposition; it is one operational realization of crossing, not a universal graph-theoretic definition.

- **S2: Connectedness** — The family-era occupancy graph is connected: there is a path between any two families through shared eras. Connectedness ensures that the population forms a single comparison component rather than disconnected subpopulations.
- **S3: Full rank** — $\mathrm{rank}(X) = p$ where X is the fitted design matrix with $p = 1 + (F-1) + (E-1)$ columns. Full column rank makes the fixed-effect coefficients uniquely determined.

Numerical Stability Gate (N1–N2):

- **N1: Conditioning** — $\kappa(X^\top X) \leq \kappa_{\max} = 100$. The bound prevents variance-component estimates from being dominated by numerical amplification of small perturbations.
- **N2: Variance inflation** — $\max(\mathrm{VIF}) \leq \mathrm{VIF}_{\max} = 10$. The bound prevents multicollinearity from inflating estimation uncertainty beyond acceptable bounds.

Failure of any gate condition means the decomposition is not reportable:

$$\boxed{\text{PASS} \iff S1 \wedge S2 \wedge S3 \wedge N1 \wedge N2}$$

The thresholds are pre-specified and validated through simulation; they are not universal constants.

V. GATED RESEARCH PROTOCOL

A. PROTOCOL ARCHITECTURE

The protocol proceeds in stages, each with an exit gate. The key design principle is identifiability before results, simulation before real data, and study-population design before empirical inference.

Stage 1: Structural population audit. Assemble the candidate model population from documented release metadata. Verify the family-era incidence is crossed and connected. Check the structural gate conditions (S1–S3).

Stage 2: Simulation estimator validation. Before any real data are analyzed, validate the estimator on simulated data with known ground truth under three regimes: balanced crossed (D1), realistic occupancy (D2), and nested/non-identifiable (D3). The estimator must recover ground truth under D1–D2 and fail detectably under D3.

Stage 3: Outcome-independent population selection. Select the smallest identifiable, recoverable subpopulation without reading any trait values. This procedure uses only occupancy metadata, lineage edges, identifiability constraints, and cost—never model accuracy, trait values, or evaluation outputs.

Stage 4: Actual model measurement. Evaluate the selected population on the target benchmark.

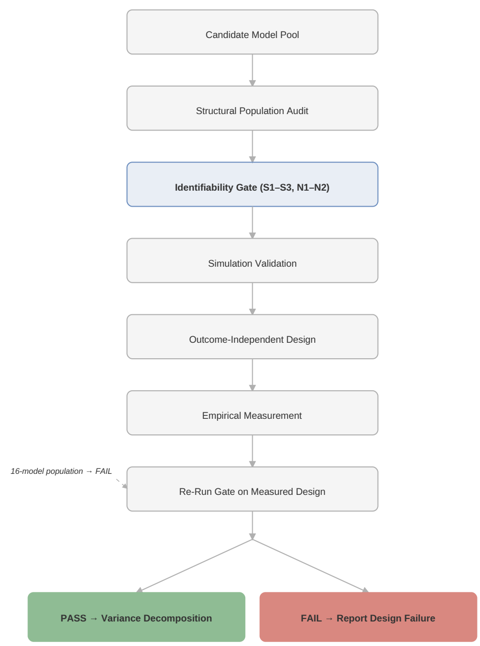

FIGURE 1: Identifiability-gated variance decomposition framework. The workflow proceeds through structural audit, simulation validation, outcome-independent population selection, measurement, and inference. Each stage has an exit gate that may halt the pipeline before empirical analysis is attempted.

Stage 5: Re-run identifiability gate on measured design. Apply S1–S3 and N1–N2 to the occupancy that the measurement pass actually returns, not the planned one. Hard failure aborts the analysis.

Stage 6: Inference only if gate passes. Estimate θ_P, compute confidence intervals, and run sensitivity analyses. If the gate fails, report the failure diagnosis without interpreting variance shares.

B. OUTCOME-INDEPENDENCE INVARIANT

The population-selection stage (Stage 3) must never read model accuracy or trait values, prediction outputs, error vectors, or evaluation metrics. It may use only family metadata, era (release quarter) metadata, lineage metadata (verified parent-offspring edges), cost estimates, and hard structural constraints (all families present, all quarters retained, edge endpoints retained). This property ensures that the resulting population can be published before measurement without creating a researcher degree of freedom.

C. ALGORITHM

Algorithm 1 states the outcome-independent population design procedure.

The selection acceptance criterion applies to era-share bias, not family-share bias. This is because era share is the limiting component under the pre-specified design: family-share bias

Algorithm 1 Outcome-independent study-population design

Input: model set $\mathcal{M}$ ($|\mathcal{M}| = N$); occupancy O (family $\times$ quarter); verified lineage edges L; per-model cost c; thresholds $\tau_{\text{bias}} = 5.0$pp (era-share bias, search) / 4.0pp (confirmation), $\tau_{\text{cov}} = 90\%$, margin $\delta = 1$pp
Assert: trait values $\mathbf{y}$ are not readable in this scope
$\mathcal{S} \leftarrow \{S \subseteq \mathcal{M} : S$ satisfies hard constraints$\}$
    All F families present in S
    Every quarter of the era window retains ≥ 1 model
    Both endpoints of every edge in L retained
    The documented within-family chain retained
For $S \in \mathcal{S}$ sorted by cost ascending:
    Build C_S; if $\text{rank}(C_S) < F + E - 1$ then continue
    If $\max(\text{VIF}(C_S)) > 10$ then continue
    If S has an era with < 2 families or a family with < 2 eras then continue
    Screen$(S, R = 300)$: if $|b_{\text{era}}| > \tau_{\text{bias}}$ or $\text{cov}_{\text{era}} < \tau_{\text{cov}}$ then continue
    Confirm$(S, R = 1000)$: if $\tau_{\text{bias}} - |b| < \delta$ then continue
    Robust$(S, R = 2000)$: if fails then continue
    For each $m \in S$: reason$[m] \leftarrow$ first constraint that fails on $S \setminus \{m\}$
    return S, reason
return $\varnothing$ (no admissible population; gate blocks measurement)

of -5.3pp in the realistic D2 scenario A is the documented six-family small-sample limit ($df = 5$) and is reported as a bound, not used as the acceptance criterion. The two-stage procedure uses a 5pp bar at search precision (300 repetitions) and tightens to a 4pp bar at confirmation precision (1000 repetitions) with a 1pp margin. Algorithm 1 checks three structural conditions (rank, VIF, and crossing) and a simulation-based recovery criterion; the condition number κ is not checked during selection and is instead reported as a post-selection diagnostic. A population that passes the selection procedure may still fail the numerical stability gate (N1) at the post-measurement audit.

VI. SIMULATION VALIDATION

A. DESIGN

Three simulation regimes (TABLE III) test the estimator under controlled conditions:

D1: Balanced crossed reference. 30 families $\times$ 14 eras $\times$ 2 models per cell (840 models). The large family count isolates estimator calibration from the small-sample limit of the real design. True components: $\sigma_L^2 = 0.5$, $\sigma_E^2 = 0.2$, $\sigma_U^2 = 0.3$ (lineage-dominant scenario A); $\sigma_L^2 = 0.2$, $\sigma_E^2 = 0.5$, $\sigma_U^2 = 0.3$ (era-dominant scenario B); $\sigma_L^2 = 0.33$, $\sigma_E^2 = 0.33$, $\sigma_U^2 = 0.34$ (balanced scenario C).

D2: Realistic occupancy. The occupancy matrix copied from the 47-model candidate population (6 families $\times$ 14 calendar quarters, sparse cells; only 11 quarters are occupied in the observed 16-model population). Same true components as D1. Bias or collapse here is a gate failure.

D3: Nested non-identifiable reference. Each family confined to a single era, so σ_L^2 and σ_E^2 are perfectly aliased. Three complementary detectors must flag the aliasing.

All regimes use 300 repetitions per scenario on recorded seeds. The estimator is the direct REML maximizer, using the Woodbury identity for computational efficiency.

TABLE III: Simulation Regimes.

Regime	Families	Eras	N	Description
D1	30	14	840	Balanced crossed reference
D2	6	14	47	Realistic sparse occupancy
D3	6	6	72	Nested non-identifiable ref.

B. D1: BALANCED CROSSED RECOVERY

TABLE IV summarizes D1 and D2 results (Fig. 2). The estimator recovers known ground truth to within 2.5 percentage points under balanced occupancy. Coverage is 88–96% across components and scenarios. All 300 repetitions converge without warnings.

C. D2: REALISTIC OCCUPANCY RECOVERY

Family-share bias is within 5.3 percentage points under realistic occupancy. The -5.3pp family-share bias in scenario A is the documented six-family small-sample limit ($df = 5$) and is reported as a limit, not corrected away. Coverage is 95–100% across components, with the wider confidence intervals reflecting the sparse cell structure.

D. D3: NESTED DESIGN DETECTION

TABLE V and Fig. 3 show that all three detectors flag the aliasing in 100% of repetitions with zero silent coverage. A detector that accepts the nested design would constitute a false-negative identification test. The threshold 1.9207 is the χ_1^2 95% critical value halved. Because the nested design produces maximally wide confidence intervals, the per-rep CI coverage of the true variance share is 16–27% (scenario-dependent); this is a consequence of the non-identifiability, not evidence of estimator calibration.

E. D4: SMALL-N STRESS TEST

TABLE VI reports REML recovery at population sizes from $N = 10$ to $N = 30$ under the realistic D2 occupancy (scenario A, 100 repetitions per size). The "Recovery Gate" column reports whether the estimator recovers the true variance share within the bias and coverage criteria from the D1–D2 simulations; this is distinct from the identifiability gate (S1–S3, N1–N2), which applies to the design structure. Under the synthetic D2 assignment used in this stress test, the recovery criterion is satisfied at $N \geq 16$ when all 6 families are represented, except at $N = 12$–14: $N = 12$ fails due to coverage (80%, below the 90% threshold), while $N = 14$ fails due to bias (+6.82pp, exceeding the bias bar) despite acceptable coverage (98%). At $N = 10$, the recovery criterion is met despite the compressed temporal structure; this is a sensitivity artifact rather than evidence of reliable estimation. This does not imply that the actual 16-model empirical population is sufficient: the empirical design fails the identifiability gate due to structural rank deficiency, whereas the synthetic D4 designs at $N \geq 16$ have different occupancy structures that avoid this particular confound. The

TABLE IV: Simulation Recovery and Failure Results. D1 balanced-crossed (300 repetitions, 30 families $\times$ 14 eras $\times$ 2 models) and D2 realistic occupancy (300 repetitions, 47-model occupancy). Family-share bias $(\hat{s}_{\text{family}} - s_{\text{family}}) \times 100$ pp is reported as signed mean over converged repetitions; coverage is 95% CI coverage.

Scenario	D1: Balanced		D2: Realistic		Conv.
	Bias (pp)	Cov.	Bias (pp)	Cov.	
A (lineage-dom.)	$+0.19$	94%	-5.34	100%	100%
B (era-dom.)	$+1.24$	88%	-0.63	99%	100%
C (balanced)	-0.30	94%	-3.46	100%	100%

TABLE V: D3: Failure-Detection Evaluation (300 repetitions, nested mis-specification).

Detector	Threshold	Sc. A	Sc. B
BLUP collinearity ($\|r\| > 0.9$)	> 0.9	100%	100%
SE inflation (SE $\geq \|\hat{\sigma}\|$)	ratio ≥ 1	100%	100%
Profile flatness (< 1.9207)	< 1.9207	100%	100%
Silent CI coverage	$= 0\%$	0%	0%

D4 results (Fig. 4) show that REML recovery is feasible at small N under favorable occupancy, but only when the design satisfies the identifiability requirements.

F. LIABILITY TEST: BINARY VERSUS CONTINUOUS TRAITS

A liability decision was resolved before the trait path was fixed. On item-level binary outcomes at the realistic 47-model occupancy, both a linear-probability model and a binomial GLMM under-estimate the era component and drive it to the boundary in 20–60% of repetitions. Continuous per-model traits (model accuracy proportions) recover the era component with 98–100% coverage. The Phase 2 estimator path is therefore LPM-REML on continuous per-model traits, not raw item-level binary responses.

G. FAMILY-ERA INTERACTION

A family-era interaction term was simulated and documented as non-identified at the sparse occupancy (SE/estimate ratios $\approx 10^4$). The interaction is excluded by design rather than absorbed.

H. IMPLICATIONS FOR THE ESTIMATION PATH

The simulation results support the continuous-trait REML specification and establish the recovery limitations that are incorporated into the population-selection criterion: (1) the continuous-trait path is preferred over binary item-level responses, and (2) the six-family small-sample limit on family-share coverage is disclosed as a known boundary condition.

VII. EMPIRICAL DATASET AND EVALUATION PROTOCOL

A. POPULATION CONSTRUCTION TRANSPARENCY

Following the transparency practice of large-scale empirical studies [1], we document every transition in our population-construction procedure. The candidate population comprises 47 open-weight models spanning 6 families (Llama, Qwen, DeepSeek, Mistral, Phi, Gemma) and 14 release quarters within 2023Q1–2026Q2. The outcome-independent selection procedure (Algorithm 1) reduces this to a 22-model candidate subset satisfying all pre-specified structural requirements (full family-era crossing, connectedness, adequate family count, and recovery-criterion calibration). Of the 22 planned models, 16 were evaluated under the available compute budget; DeepSeek-V3.1 and DeepSeek-V3.2 could not be evaluated due to compute constraints, and four additional models were deprioritized during the evaluation campaign. The transitions are:

$$\underbrace{47}_{\text{candidate}} \longrightarrow \underbrace{22}_{\text{selected}} \longrightarrow \underbrace{16}_{\text{evaluated}}$$

Every model in the candidate population has documented family metadata (verified against Hugging Face organization metadata and technical reports) and release quarter (public release date, not the HuggingFace `createdAt` timestamp; documented divergences corrected by hand). The metadata fields recorded for each model are: model name, family, release date, release quarter, parameter count, access type (public or gated), evaluation source, and measurement status.

B. MEASURED POPULATION

Sixteen of the planned models were evaluated under the available compute budget, within the 47-model candidate frame. The 16 models span 5 families (Llama, Qwen, Mistral, Phi, Gemma) and 11 occupied release quarters within the 2023Q1–2026Q2 observation window. Not all planned models could be evaluated: DeepSeek-V3.1 and DeepSeek-V3.2 were not evaluated due to compute constraints.

C. BENCHMARK

All evaluations use MMLU [30] with 5-shot prompting and 14,042 evaluation samples per model. Results are aggregate CSV accuracy values.

D. MODEL METADATA

Each model is characterized by: family (verified against Hugging Face organization metadata and technical reports); release quarter (public release date, not the HuggingFace `createdAt` timestamp; documented divergences corrected by hand); repository and version identifier; parameter count; access type (public or gated); evaluation source (direct evaluation or leader board); and measurement status (evaluated or unevaluated).

E. FIDELITY

Fifteen models were evaluated at BF16 precision. One model (Mistral-Small-4, 119B parameters) was evaluated at 4-bit quantization due to memory constraints. The fidelity column in TABLE VII records this distinction.

TABLE VI: D4: Small-N Recovery Results. REML recovery under D2 occupancy, scenario A (100 repetitions per size). "Recovery Gate" reports whether bias and coverage criteria are met; this is distinct from the structural/numerical identifiability gate (S1–S3, N1–N2). Results are specific to the synthetic D2 assignment used here.

N	F	E	Rank	κ	Bias (pp)	Cov.%	Rec. Gate
10	5	5	9/9	158	$+4.84$	97	PASS
12	5	8	12/12	89	$+4.64$	80	FAIL
14	5	10	14/14	52	$+6.82$	98	FAIL
16	6	10	15/15	47	$+1.49$	98	PASS
18	6	12	17/17	42	$+1.56$	96	PASS
20	6	10	15/15	38	$+2.25$	97	PASS
22	6	10	15/15	31	-0.36	98	PASS
24	6	10	15/15	28	$+0.43$	99	PASS
30	6	13	18/18	22	$+3.41$	96	PASS

F. HARDWARE

Evaluations were performed on rented cloud GPUs, including NVIDIA A100-SXM4 80GB configurations. Of the 16 measured models, 15 were evaluated at BF16 precision and 1 (Mistral-Small-4, 119B) was evaluated at 4-bit quantization for memory efficiency.

G. EVALUATION VALIDITY CAVEAT

Six models produce accuracy values near the chance level for a 4-choice benchmark (23–25%): Devstral-2 (25.1%), Phi-1 (24.8%), Mistral-Small-4 (24.3%), Mistral-Small-3.1 (23.4%), Mistral-Small-3.2 (23.1%), and Qwen-7B (22.9%). The Mistral-Small family exhibits a pronounced discontinuity: Mistral-Small-3 (Jan 2025) scores 80.7%, while Mistral-Small-3.1 (Mar 2025) and Mistral-Small-3.2 (Jun 2025)—the same 24B architecture, two and five months later—score 23.4% and 23.1% respectively.

These results trigger an evaluation-validity audit. Until tokenizer configuration, prompt formatting, answer-extraction logic, and benchmark configuration are independently verified for each model, the corresponding aggregate accuracies should be treated as provisional. The identifiability gate depends only on the design matrix structure (which families and eras are represented), not on the trait values themselves. The gate result—rank 14 of 15, $\kappa = 4.7 \times 10^{16}$, VIF $= \infty$—is invariant to the accuracy values. However, any future study that intends to interpret the variance-component estimates must first validate the accuracy measurements against independent reference scores.

H. AGGREGATE ACCURACY

Status: provisional. The following aggregate accuracies are reported for descriptive and diagnostic context only; their evaluation validity is pending independent verification (Sec. XI). They are *not* interpreted as validated benchmark measurements. TABLE VII lists the 16-model empirical accuracy on MMLU (5-shot, 14,042 items). Fig. 5 shows the distribution.

I. DATA INTEGRITY VALIDATION

The per-question JSONL prediction files were validated for all 16 models. All 16 files contain identical constant predictions: every model predicts answer 0 for every question. This confirms that the JSONL artifacts are non-informative and inconsistent with the reported aggregate evaluation; they cannot be treated as actual per-question model predictions. The CSV-level aggregate accuracy values were used as inputs to the diagnostic (not inferential) REML fit in Section IX; their evaluation validity remains provisional per Section VII and does not affect the gate diagnostics in Section VIII, which depend only on design structure. The error-similarity analysis specified in the original plan cannot be performed on this dataset and is deferred to future work once validated per-question outputs are available.

J. WHAT IS AND IS NOT MEASURED

The following populations exist in the project but are NOT measured empirical results:

- **47-model candidate population:** The full structural population from the Phase 0 audit. Not measured.
- **22-model selected population:** The candidate subset within the 47-model candidate frame identified by the outcome-independent design procedure. Not fully measured (16 of the planned models were evaluated; DeepSeek-V3.1/V3.2 could not be evaluated due to compute constraints).
- **DeepSeek models:** DeepSeek-V3.1 (671B) and DeepSeek-V3.2 (685B) were NOT empirically measured. They must not be described as measured results.

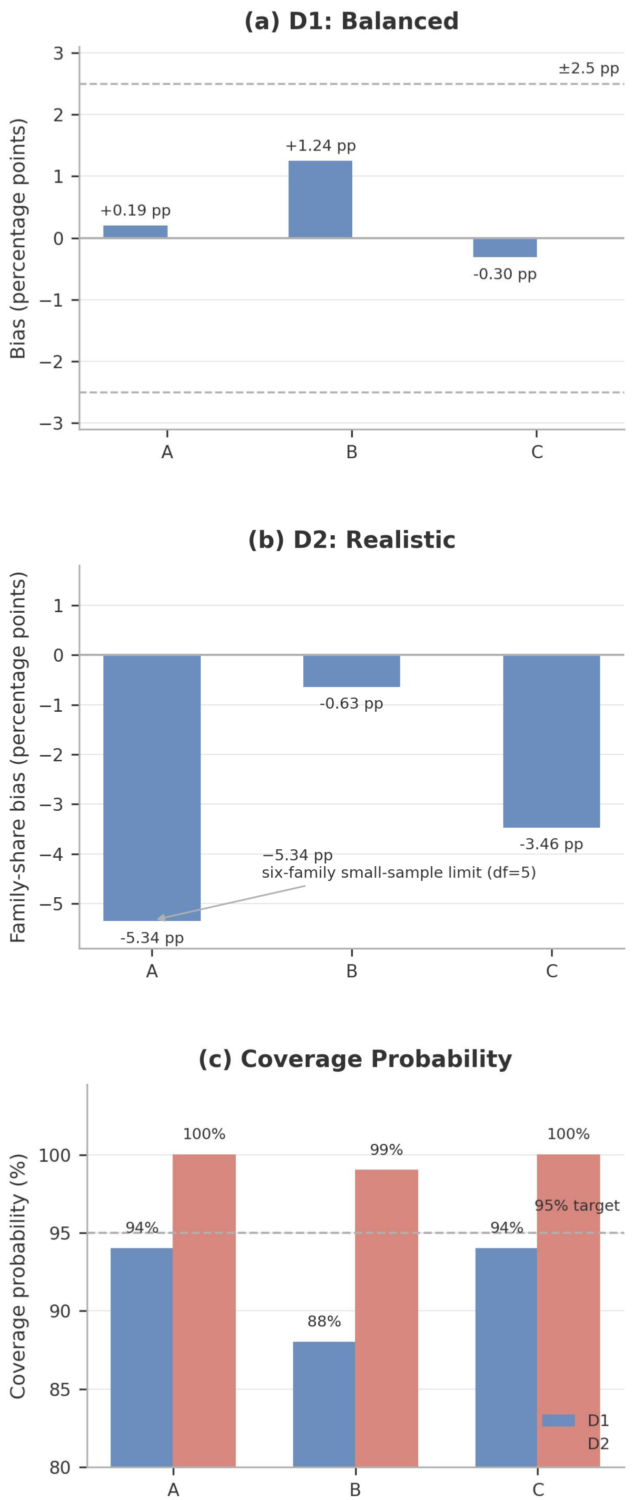

FIGURE 2: Simulation recovery across D1 (balanced) and D2 (realistic) regimes. (a) D1 family-share bias; (b) D2 family-share bias; (c) 95% CI coverage. Each bias bar reports the signed family-share bias $(\hat{s}_{\text{family}} - s_{\text{family}}) \times 100\,\text{pp}$, averaged over converged repetitions. The -5.3pp family-share bias in scenario A under D2 is the documented six-family small-sample limit (df $= 5$).

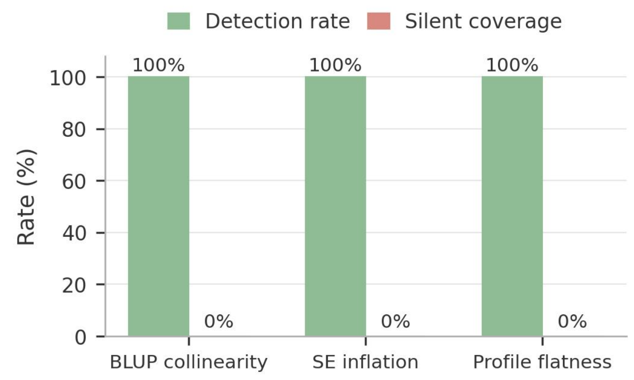

FIGURE 3: D3 failure detection. Three complementary failure detectors (BLUP collinearity, SE inflation, profile flatness) flag nested aliasing in 100% of repetitions with zero silent coverage across all scenarios.

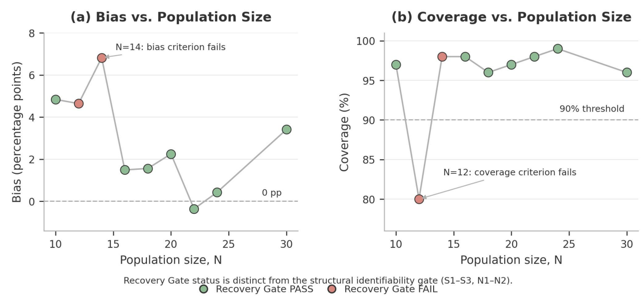

FIGURE 4: D4 small-N stress test. REML recovery (bias and 95% CI coverage) at population sizes $N = 10$–30 under the realistic D2 occupancy (scenario A, 100 repetitions per size). Shaded regions indicate the bias and coverage thresholds. $N = 12$–14 fail the recovery criterion despite satisfying the identifiability gate.

VIII. EMPIRICAL IDENTIFIABILITY RESULTS

A. FAMILY-ERA OCCUPANCY

TABLE VIII shows the family-era occupancy of the 16 measured models. Three quarters are unoccupied (2024Q1, 2024Q3, 2025Q3). Eight of 11 occupied quarters contain only one model. No measured models belong to the DeepSeek family. The occupancy matrix is sparse: 40 of 55 possible family-quarter cells are empty (73% sparse). The full family-era occupancy is shown in Fig. 6.

B. GATE DIAGNOSTICS

The observed 16-model population spans 11 occupied release quarters within a 14-quarter observation window (2024Q1, 2024Q3, and 2025Q3 are unoccupied). After removing unoccupied factor levels, the fitted design matrix has $p = 1 + (F - 1) + (E - 1) = 1 + 4 + 10 = 15$ columns. TABLE IX and Fig. 7 summarize gate diagnostic results. Four of five gate conditions fail; connectedness (S2) passes.

C. FAILURE DIAGNOSIS

The fitted 16-model design contains five observed families (Llama, Qwen, Mistral, Phi, Gemma) and eleven occupied

TABLE VII: Aggregate MMLU Measurements Used for Descriptive and Diagnostic Analysis (5-shot, 14,042 items). 16 of the planned models were evaluated. Six models produce accuracy near chance level (23–25%) and are flagged for independent audit. All measurements are provisional pending evaluation-validity verification.

Model	Family	Era	Acc.	Params	Fidelity	Source	Validity
Mistral-Small-3	Mistral	2025Q1	0.8069	24B	BF16	Direct	Provisional
Phi-3	Phi	2024Q2	0.7799	14B	BF16	Direct	Provisional
Phi-4-reas.+	Phi	2025Q2	0.7782	14B	BF16	Direct	Provisional
Phi-4	Phi	2024Q4	0.6864	14B	BF16	Direct	Provisional
Gemma-3n	Gemma	2025Q2	0.6364	4B	BF16	Direct	Provisional
Mistral-7B	Mistral	2023Q3	0.6186	7B	BF16	Direct	Provisional
Phi-2	Phi	2023Q4	0.5644	2.7B	BF16	Direct	Provisional
Gemma-4-12B	Gemma	2026Q2	0.4397	12B	BF16	Direct	Provisional
Phi-1.5	Phi	2023Q3	0.4218	1.3B	BF16	Direct	Provisional
Llama-1	Llama	2023Q1	0.3424	7B	BF16	Direct	Provisional
Devstral-2	Mistral	2025Q4	0.2515	24B	BF16	Direct	**Audit req.**
Mistral-Small-3.1	Mistral	2025Q1	0.2340	24B	BF16	Direct	**Audit req.**
Phi-1	Phi	2023Q2	0.2480	1.3B	BF16	Direct	**Audit req.**
Mistral-Small-4	Mistral	2026Q1	0.2433	119B	4-bit	Direct	**Audit req.**
Mistral-Small-3.2	Mistral	2025Q2	0.2314	24B	BF16	Direct	**Audit req.**
Qwen-7B	Qwen	2023Q3	0.2295	7B	BF16	Direct	**Audit req.**

TABLE VIII: Family-Era Occupancy of the 16 Measured Models. 8 of 11 occupied quarters contain only one model; 40 of 55 possible family-quarter cells are empty (73% sparse).

Fam.	2023				2024		2025			2026	
	Q1	Q2	Q3	Q4	Q2	Q4	Q1	Q2	Q4	Q1	Q2
Llama	1										
Qwen			1								
Mistral			1				2	1	1	1	
Phi		1	1	1	1	1		1			
Gemma								1			1

TABLE IX: Identifiability Gate Audit for the 16-Model Population. Four of five conditions fail. The occupancy graph remains connected (S2 passes) but Llama and Qwen each occupy a single era, violating the crossing requirement (S1).

Gate	Criterion	Observed	Threshold	Status
S1	Crossing	Llama, Qwen in 1 era	Every family ≥ 2 eras	**FAIL**
S2	Connectedness	Connected graph	Connected	PASS
S3	Rank	14	$= 15$	**FAIL**
N1	Condition number κ	4.72×10^{16}	≤ 100	**FAIL**
N2	Max VIF	∞	≤ 10	**FAIL**

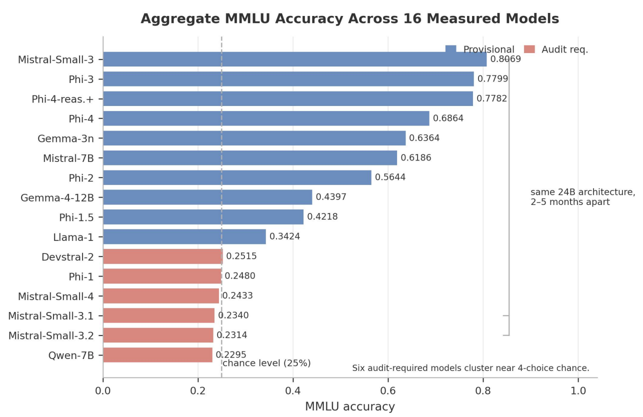

FIGURE 5: Distribution of measured MMLU accuracy (provisional). These are descriptive aggregate values reported for diagnostic context only; their evaluation validity is pending an independent audit because the recorded per-question answers were non-informative (Sec. XI). Six models cluster near the chance level (23–25%) for a 4-choice benchmark, triggering an evaluation-validity audit. The identifiability gate is invariant to accuracy values.

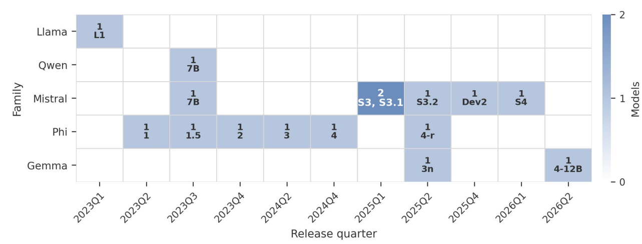

FIGURE 6: Family×era occupancy heatmap of the 16 measured models. Darker shading indicates more models in a cell. 8 of 11 occupied quarters contain only one model; 40 of 55 possible cells are empty (73% sparse).

release quarters, giving $p = 1+4+10 = 15$ columns. A null-space computation on the fitted design matrix shows that the rank deficiency (rank 14 of 15) is fully explained by a single exact linear dependency: the Llama family-dummy column equals the 2023Q1 era-dummy column, because Llama-1 is the only model in that quarter. The null-space vector is $(\ldots, +1, \ldots, -1, \ldots)$ with nonzero entries at the Llama and 2023Q1 positions.

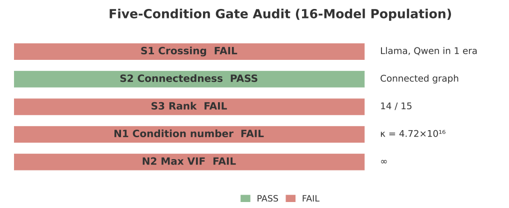

FIGURE 7: Gate diagnostics versus pre-specified thresholds. The 16-model population fails four of five gate conditions: S1 (crossing) fails because Llama and Qwen each occupy a single era; S3 (rank) fails at 14/15; N1 (conditioning) fails at $\kappa = 4.7 \times 10^{16}$; N2 (VIF) fails at ∞. S2 (connectedness) passes: the occupancy graph remains connected.

Qwen also occupies a single quarter (2023Q3), which violates the crossing requirement (each family must span ≥ 2 quarters), but this does not contribute to the rank deficiency: the Qwen family-dummy column (one 1, at Qwen-7B) differs from the 2023Q3 era-dummy column (three 1s, at Qwen-7B, Mistral-7B, and Phi-1.5) and is linearly independent of it. The crossing violation is a separate structural deficiency that the gate flags but is not the proximate cause of the rank collapse.

IX. CONSEQUENCES OF IGNORING THE GATE

A. DIAGNOSTIC ESTIMATES UNDER INVALID DESIGN

If the identifiability gate is not applied, the REML estimator converges and produces variance-component estimates:

- $\hat{\sigma}^2_{\text{family}} = 0.00274$
- $\hat{\sigma}^2_{\text{era}} \approx 2.06 \times 10^{-9}$
- $\hat{\sigma}^2_{\text{unique}} = 0.04778$
- Point-estimate family share: 5.4%

These estimates must be interpreted as diagnostic restricted-design estimates, not as inferential results. The near-zero era estimate is not evidence of no era effect. It is an aliasing artifact of the rank-deficient design, where era variance cannot be separated from residual variance.

B. CONFIDENCE INTERVALS

The 95% confidence interval for the family share covers $[0\%, 100\%]$ (see Appendix A for the diagnostic variance-component plot). The bootstrap 95% interval is $[3.4 \times 10^{-8}, 0.776]$—extremely wide and practically uninformative. Leave-one-model-out analysis shows the estimate is driven by 2–3 influential observations: removing Mistral-Small-3 alone shifts the family share from 5.4% to 26.4%, and removing any of 7 other models collapses it to 0%.

C. WHY CONVERGENCE DOES NOT IMPLY IDENTIFIABILITY

A numerical optimizer (Nelder-Mead) can converge to a finite optimum of the restricted log-likelihood even when the design is rank-deficient. The likelihood surface has a ridge along the aliased direction, and the optimizer's stopping point on that ridge depends on the starting value and implementation details. The resulting estimate is well defined only relative to the software. The gate prevents inference under this failure mode: it catches the non-identifiability before anyone interprets the 5.4% number.

X. POPULATION-DESIGN SENSITIVITY ANALYSIS

A. DESIGN SPACE

We systematically sweep alternative population designs to characterize which configurations satisfy the identifiability requirements. The design parameter space covers 5–7 families, 8–14 eras, and 2–6 models per family, with staggered era assignments that enforce family-era crossing and connectedness by construction. The sweep evaluated 105 configurations in total (3 family counts $\times$ 7 era counts $\times$ 5 models-per-family values, each producing a single deterministic staggered assignment). Of these, 40 achieve full rank, 3 pass the condition-number gate ($\kappa \leq 100$), and 3 satisfy all three numerical/design-matrix diagnostics (rank, κ, VIF). The verified passing configuration has $N = 30$ (6 families $\times$ 5 each, 8 eras); its rank, condition number, and VIF are reported in TABLE X.

B. RESULTS

TABLE X summarizes selected population designs and their identifiability status. The "Manual (20)" row is a hand-constructed 20-model comparison design included as a diagnostic reference; unlike the deterministic uniform-(M) sweep, it uses a non-uniform family allocation across 6 families and 14 eras.

C. KEY FINDINGS

Figs. 8–11 visualize the design-space sweep.

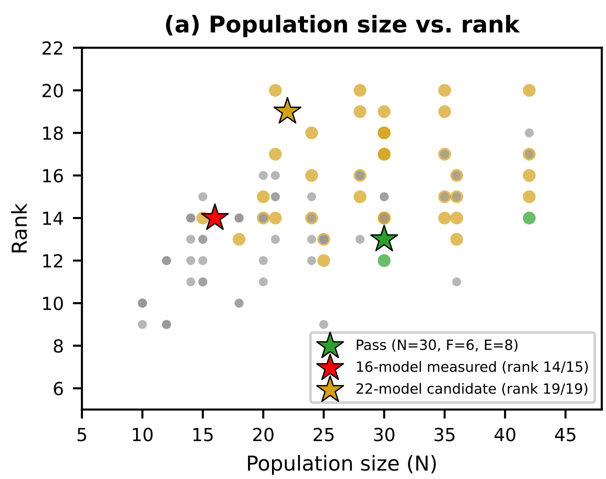

FIGURE 8: Population size versus rank across the design space. Each point is one evaluated (F, E, M) configuration. The passing design (N = 30, F = 6, E = 8) is highlighted; the measured 16-model population (rank 14/15) and the 22-model candidate (rank 19/19) are marked as reference points.

Full rank alone is insufficient. The 6-families $\times$ 3-each design achieves full rank (rank 13/13) but fails the condition

TABLE X: Design-Space Sensitivity Analysis. Selected population designs and their identifiability status. 30 models across 6 families and 8 eras achieves rank 13/13, $\kappa = 93$, and VIF $= 2.1$—the smallest observed passing configuration in this design sweep, not the universal minimum.

Config.	N	F	E	Rank	κ	VIF	Status	Binding
Current (16)	16	5	11	14/15	4.7×10^{16}	∞	**FAIL**	S1+S3+κ+VIF
Manual (20)	20	6	14	16/19	3.5×10^{18}	10.9	**FAIL**	S3+κ+VIF
6 fam×3, 8 eras	18	6	8	13/13	164	4.7	**FAIL**	κ
6 fam×4, 10 eras	24	6	10	13/15	∞	4.2	**FAIL**	S3+κ
6 fam×5, 8 eras	**30**	**6**	**8**	**13/13**	**93.0**	**2.1**	**PASS**	—
6 fam×5, 12 eras	30	6	12	17/17	204	3.2	**FAIL**	κ

number gate ($\kappa = 164 > 100$). Numerical conditioning is an additional necessary requirement beyond rank.

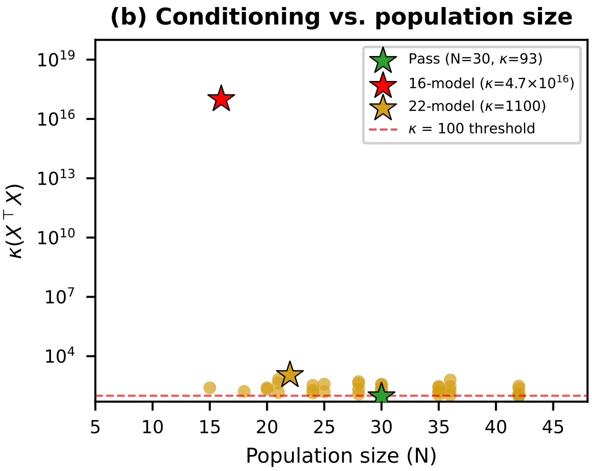

FIGURE 9: Conditioning versus population size. The condition number κ is shown for all full-rank designs, with the $\kappa = 100$ numerical-gate threshold marked. The passing design (N = 30, $\kappa = 93$) sits below the threshold; the measured 16-model population ($\kappa = 4.7 \times 10^{16}$) and the 22-model candidate ($\kappa = 1100$) are noted as reference points.

Increasing eras while holding N fixed worsens conditioning. The 30-model design with 8 eras passes ($\kappa = 93$), but the same 30 models across 12 eras fails ($\kappa = 204$). More era columns relative to rows increase multicollinearity among era indicators.

The joint κ–VIF partition in Fig. 10 shows that no full-rank design outside the $N \geq 25$ range simultaneously satisfies both numerical thresholds. Fig. 11 confirms this at the population level: among 105 evaluated designs, only three pass all five gate conditions.

One sufficient configuration. 30 models across 6 families and 8 eras, with balanced occupancy (five models per family and approximately 3–4 models per era), achieves rank 13/13, $\kappa = 93$, and max VIF $= 2.1$. This is one sufficient design configuration within the evaluated grid—smaller or differently structured populations may also pass.

D. OUTCOME-INDEPENDENT SELECTION WITHIN THE CANDIDATE FRAME

The selection procedure (Algorithm 1) selects the candidate subset from the 47-model candidate set. The structural minimum is 21 models (identifiable: full rank, VIF $\leq$ 10). This minimum is derived by the procedure rather than assumed: it is the smallest population count for which any outcome-independent, connected subset of the 47-model candidate frame induces a family-by-era design that is simultaneously full-rank and VIF-bounded, subject to the per-family span and per-quarter occupancy constraints—below 21 models those structural constraints cannot all be met within this candidate frame. The 21-model design, however, sits exactly on the recovery bar in simulation (era-share bias ≈ -5.0pp with SE 0.6–0.8pp—a knife-edge). Within the 47-model candidate frame, the selection procedure identifies a **22-model candidate subset** that clears the recovery bar at 300 repetitions (era bias A $+2.4$pp / B -0.8pp), passes the 1000-repetition margin confirmation (A $+2.2$pp / B -2.4pp), and is robust at 2000 repetitions (A $+1.7$pp / B -3.2pp). The extra model over the structural minimum is required by the pre-specified recovery criterion within this candidate frame. The 22-model population satisfies the structural gate conditions (S1–S3, N2) but does not satisfy the condition-number gate ($\kappa = 1100 > 100$); Algorithm 1 checks rank, VIF, and crossing but does not check κ, which is instead reported as a post-selection diagnostic. The 22-model subset is therefore not a fully gate-passing population; it is the outcome-independent candidate within this specific frame and under these specific selection criteria.

E. ROBUSTNESS AND SENSITIVITY

We characterize the sensitivity of the gate results to threshold choices and population perturbations.

1) Condition-number threshold.

The baseline gate uses $\kappa_{\max} = 100$. At $\kappa_{\max} = 50$ the 16-model population still fails ($\kappa = 4.7 \times 10^{16}$), and the sufficient 30-model design still passes ($\kappa = 93$). At $\kappa_{\max} = 200$, the 6-family $\times$ 3-each design (18 models, rank 13/13, $\kappa = 164$) passes. The PASS/FAIL boundary is therefore robust across a factor-of-4 range in $\kappa_{\max}$.

2) VIF threshold.

The baseline gate uses VIF$_{\max} = 10$. No design in the sweep passes all other gates while failing the VIF threshold,

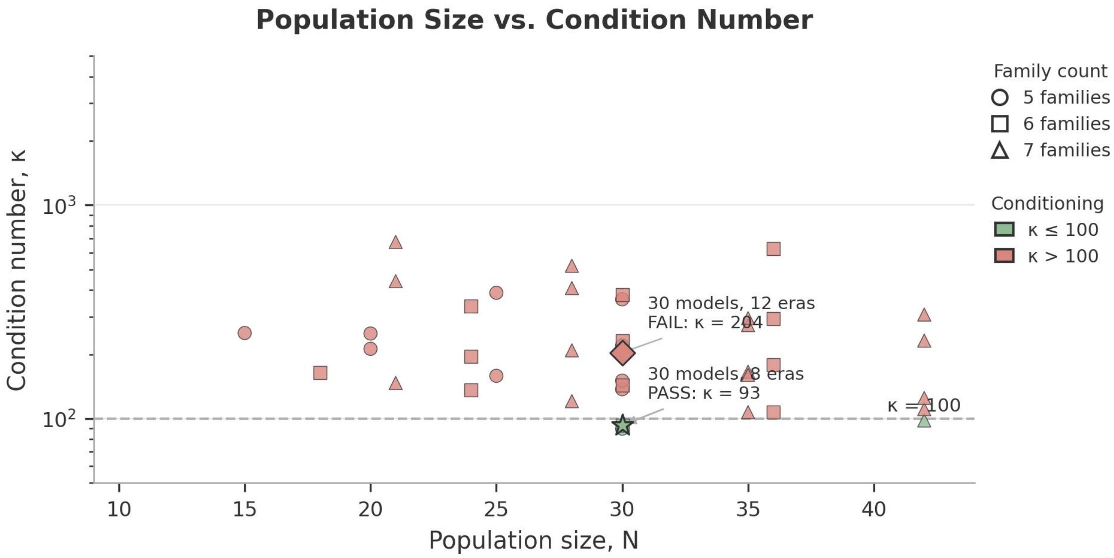

FIGURE 10: Condition number versus VIF across full-rank designs. The two numerical gate thresholds ($\kappa = 100$, VIF $= 10$) partition the design space into pass and fail regions. Passing designs (green) satisfy both thresholds simultaneously.

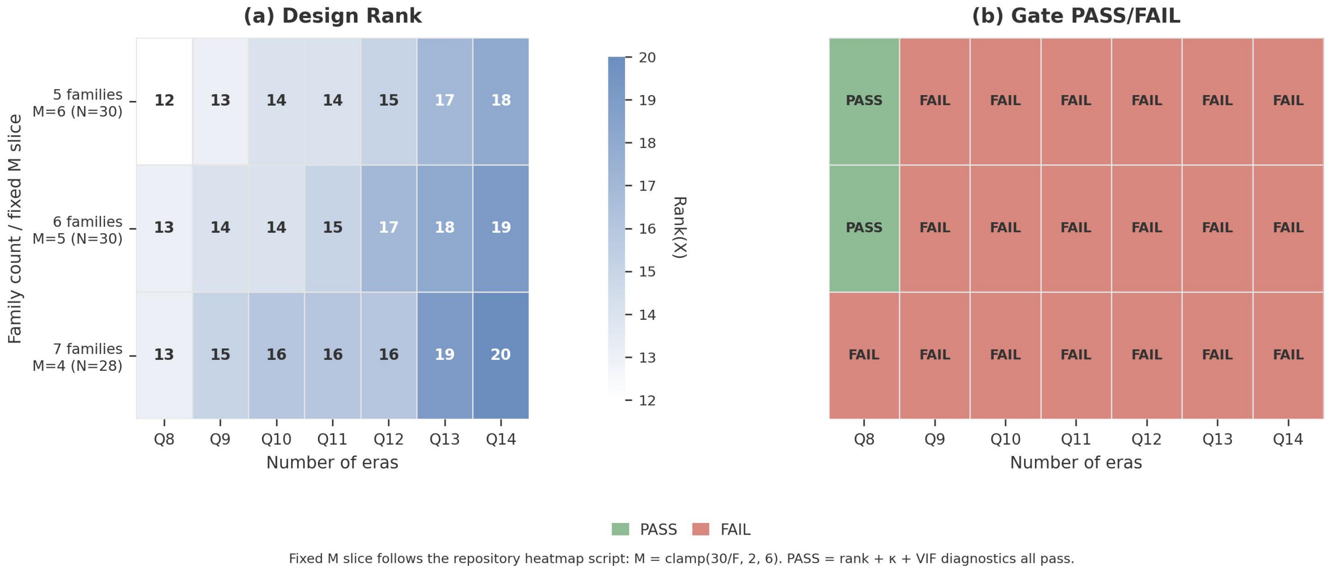

FIGURE 11: PASS/FAIL heatmap across the design space. Each cell shows whether a given (N, F, E) configuration passes all three numerical/design-matrix diagnostics: full rank, $\kappa \leq 100$, and VIF ≤ 10. Crossing and connectedness are enforced by the staggered assignment constraints.

so the VIF gate is not the binding constraint for any evaluated configuration.

3) Leave-one-model-out.

TABLE XI reports structural diagnostics for each of the 16 leave-one-model-out configurations. Removing Llama-1 alone restores full rank (13/13) by eliminating the Llama family level and the 2023Q1 era level simultaneously. However, the resulting design does not satisfy the full identifiability gate: the condition number $\kappa = 585 > 100$, so the numerical stability criterion fails. All other removals leave the design rank-deficient with $\kappa \gg 10^{16}$ (Fig. 12). These results show that rank deficiency and numerical instability are separable problems: fixing rank does not automatically fix conditioning.

4) Era granularity.

TABLE XII reports gate diagnostics at three temporal resolutions (Fig. 13). At quarter level (baseline), the design is rank-deficient with catastrophic conditioning ($\kappa = 4.7 \times 10^{16}$). Aggregating to half-year ($E = 7$ occupied levels) restores full rank (11/11) and improves conditioning to $\kappa = 227$, but

TABLE XI: Leave-One-Model-Out Structural Diagnostics. Removing Llama-1 restores full rank but does not satisfy the numerical stability gate ($\kappa = 585 > 100$). All other removals leave the design rank-deficient. The "Binding" column identifies which gate criterion remains unsatisfied after removal.

Removed	N	F	E	Rank	κ	Status	Binding
Llama-1	15	4	10	13/13	585	**FAIL**	κ
Qwen-7B	15	4	11	13/14	3.9×10^{17}	FAIL	rank + κ
Mistral-7B	15	5	11	14/15	8.2×10^{16}	FAIL	rank + κ
Mistral-Small-3	15	5	11	14/15	4.9×10^{16}	FAIL	rank + κ
Mistral-Small-3.1	15	5	11	14/15	4.9×10^{16}	FAIL	rank + κ
Mistral-Small-3.2	15	5	11	14/15	6.2×10^{16}	FAIL	rank + κ
Devstral-2	15	5	10	13/14	6.7×10^{16}	FAIL	rank + κ
Mistral-Small-4	15	5	10	13/14	6.7×10^{16}	FAIL	rank + κ
Phi-1	15	5	10	13/14	2.0×10^{17}	FAIL	rank + κ
Phi-1.5	15	5	11	14/15	3.6×10^{17}	FAIL	rank + κ
Phi-2	15	5	10	13/14	1.6×10^{17}	FAIL	rank + κ
Phi-3	15	5	10	13/14	1.6×10^{17}	FAIL	rank + κ
Phi-4	15	5	10	13/14	1.6×10^{17}	FAIL	rank + κ
Phi-4-reas.+	15	5	11	14/15	1.1×10^{17}	FAIL	rank + κ
Gemma-3n	15	5	11	13/15	5.2×10^{17}	FAIL	rank + κ
Gemma-4-12B	15	5	10	13/14	3.2×10^{17}	FAIL	rank + κ

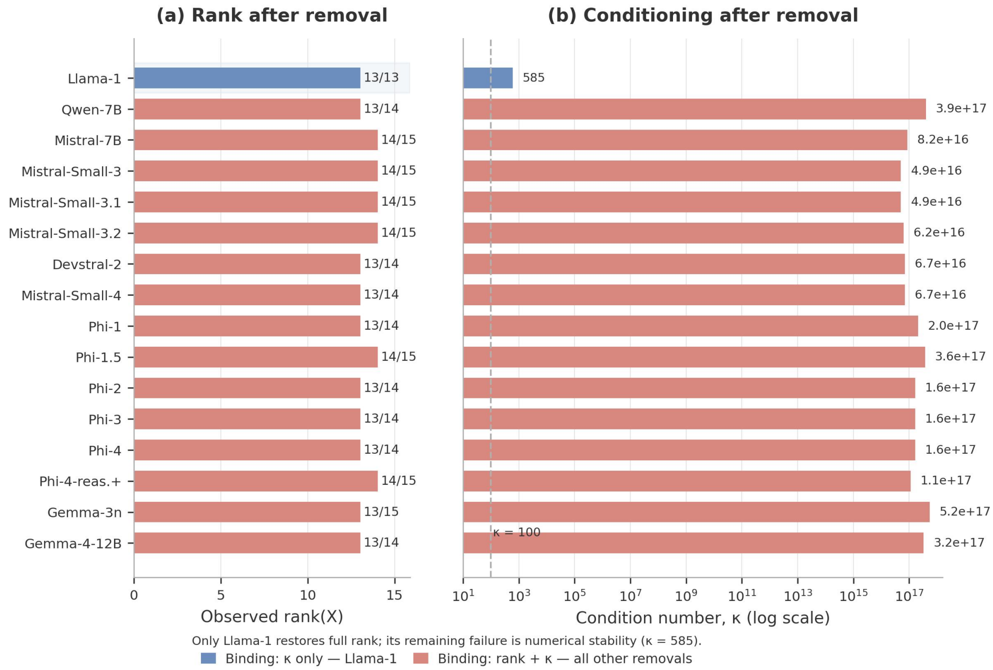

FIGURE 12: Leave-one-model-out diagnostics. Each row shows the rank and condition number after removing one model. Removing Llama-1 alone restores full rank (13/13) but $\kappa = 585 > 100$, demonstrating that rank deficiency and conditioning failure are separable problems.

the condition number still exceeds the $\kappa \leq 100$ threshold, so the numerical stability gate fails. Year-level aggregation ($E = 4$ occupied levels) also restores full rank (8/8) with $\kappa = 202 > 100$, again failing the numerical stability gate. The rank failure is specific to the quarterly resolution and is eliminated by coarser grouping. However, numerical instability persists at all three resolutions, demonstrating that rank deficiency and conditioning failure are separable problems.

5) Contrast coding.

The baseline uses treatment (reference) coding. Sum-to-zero (deviation) coding produces rank 14/15, $\kappa = 3.6 \times 10^{16}$, VIF $= \infty$—the same FAIL verdict. The rank deficiency is structural (a consequence of the Llama–2023Q1 confound) and is not an artifact of the contrast parameterization.

6) Fidelity sensitivity.

One model (Mistral-Small-4) was evaluated at 4-bit quantization rather than BF16. This does not affect the identifiability gate (which depends on design structure), but any future

TABLE XII: Era-Resolution Sensitivity. Collapsing quarters to half-years or years eliminates rank deficiency but does not satisfy the full identifiability gate: κ remains above the ≤ 100 threshold at all resolutions.

Resolution	E	p	Rank	κ	VIF	Status	Binding
Quarter (baseline)	11	15	14/15	4.7×10^{16}	∞	FAIL	rank + κ + VIF
Half-year	7	11	11/11	227	6.2	FAIL	κ
Year	4	8	8/8	202	6.0	FAIL	κ

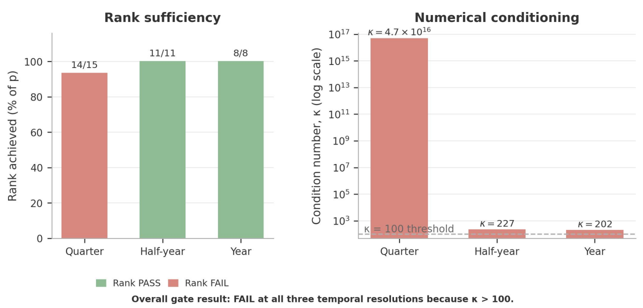

FIGURE 13: Era-resolution sensitivity. Gate diagnostics at quarter, half-year, and year resolution. Collapsing quarters eliminates rank deficiency but does not satisfy the numerical stability gate: κ remains above 100 at all resolutions.

variance-component interpretation would need to account for the fidelity confound.

XI. DATA INTEGRITY, THREATS TO VALIDITY, AND LIMITATIONS

A. DATA INTEGRITY

- The CSV aggregate accuracy values (TABLE VII) are retained for descriptive and diagnostic analysis. The identifiability gate itself depends only on population structure and is independent of the observed trait values.
- The per-question JSONL prediction files failed integrity validation: all 16 files contain constant predictions (every model predicts answer 0).
- Model-by-item error-similarity analysis is therefore not reported and is deferred to future work once validated per-question prediction artifacts are available.
- No model-by-item correlation claim is made from the empirical data.

B. INTERNAL VALIDITY

1) **Unverified evaluation validity.** Six models produce accuracy near the 4-choice chance level (23–25%), and the Mistral-Small family shows a pronounced discontinuity between versions. The CSV values record artifact completeness, not necessarily evaluation validity. The identifiability gate result is invariant to the trait values, but any future variance-component interpretation requires independent validation of the accuracy measurements.
2) **Per-question JSONL integrity failure.** The JSONL artifacts are corrupted (constant predictions). Error-similarity analysis is deferred until validated per-question outputs are available.
3) **Mixed numerical fidelity.** One model (Mistral-Small-4) was evaluated at 4-bit quantization; all others at BF16. This introduces a confound between model identity and evaluation precision.

C. CONSTRUCT VALIDITY

1) **Family $\neq$ true lineage.** The grouping factor "family" is a proxy for model lineage. True cross-generation lineage is largely undocumented in model cards; only 5 cross-generation edges are verified from metadata.
2) **Release quarter $\neq$ training era.** Release date is discretized to quarters, which simplifies temporal structure. Models released in the same quarter may have been trained on different data.
3) **MMLU accuracy $\neq$ overall capability.** The trait is measured on MMLU only; a benchmark weighted differently could yield different variance-component estimates.

D. STATISTICAL VALIDITY

1) **Small N.** With only 16 models, the design is sparse and the family-era occupancy is incomplete.
2) **Sparse family-era occupancy.** 73% of family-quarter cells are empty. Eight of 11 occupied quarters contain only one model.
3) **Threshold sensitivity.** The gate thresholds ($\kappa_{\max} = 100$, $\mathrm{VIF}_{\max} = 10$) are conventional choices validated through simulation, not universal constants.
4) **Synthetic design assumptions.** The population-design analysis assumes a specific staggered assignment strategy. Other assignment strategies may yield different conditioning.
5) **Possible benchmark contamination.** If pretraining included MMLU items, trait variance is compressed toward zero and era/lineage shares are biased downward.

E. EXTERNAL VALIDITY

1) **Open-weight only.** Conclusions are scoped to the open-weight model population and do not generalize to closed or proprietary models.
2) **Single benchmark.** Results are scoped to MMLU 5-shot accuracy. Generalizability to other benchmarks is not established.
3) **No real population passes the full gate.** The sufficient configurations identified in Section X are abstract population structures from a parameter sweep; we have not verified that a real set of currently available open-weight models instantiates this exact family-era distribution. The real-world outcome-independent candidate

(22 models, from the actual 47-model pool) does not pass the full gate ($\kappa = 1100 > 100$). No population evaluated in this study—real or synthetic—both exists today and passes all five gate conditions simultaneously.

XII. DISCUSSION

A. MAIN METHODOLOGICAL FINDING

The empirical analysis shows that variance decomposition of foundation-model performance requires pre-estimation identifiability checking, and that a realistic model population can fail those checks. Four of five gate conditions fail for the 16-model design; the 16-model design reflects the compute-feasible subset available in our study, not a deliberately pathological construction.

B. WHAT WAS AND WAS NOT DEMONSTRATED

The present study successfully demonstrates: (A) a formal identifiability framework for crossed variance-component models, (B) simulation-validated estimator behavior under identifiable and non-identifiable designs, (C) that the actual 16-model population fails the identifiability gate, and (D) a sufficient configuration within the evaluated design grid through systematic design-space analysis. The study does not demonstrate: (E) actual family-proxy-vs-era variance shares (the gate failed), (F) empirical correlated-error structure (the JSONL data are corrupted), or (G) causal lineage effects (the observational design cannot support causal claims). The distinction between (A)–(D) and (E)–(G) should be kept clear throughout.

C. INTERPRETATION OF THE NEGATIVE RESULT

The 16-model population fails four of five gate conditions. The 5.4% family-share point estimate and its $[0\%, 100\%]$ confidence interval are diagnostic outputs of an invalid design, not substantive findings about lineage effects.

D. IMPLICATIONS FOR BENCHMARK AND MODEL-PANEL DESIGN

- **For researchers:** Apply the identifiability gate before any variance decomposition of model populations. If the gate fails, the decomposition is not reportable regardless of what the estimator produces.
- **For benchmark designers:** Ensure the evaluated model population spans enough families and eras with sufficient density.
- **For the correlated-error literature:** Claims about whether diversifying across families reduces correlated error require a variance decomposition that is identifiable on the studied population.

E. WHAT ADDITIONAL EMPIRICAL DATA WOULD BE NEEDED

To perform the intended family-proxy-era variance decomposition, a future study would need:

- A population satisfying comparable structural and recoverability criteria to the 22-model selection (full family-era crossing, adequate family count, sufficient density)
- All 6 families represented with at least 2 models each
- At least 8 occupied release quarters with at least 2 families per quarter (the sufficient designs explored here generally required this density under the evaluated staggered assignment strategy)
- Validated per-question prediction data (not the corrupted JSONL artifacts available here)
- Consistent numerical fidelity across all models

XIII. REPRODUCIBILITY AND OPEN SCIENCE

All analysis code, population definitions, simulation outputs, and design artifacts are available in the project repository.

Environment: Python 3.11.7, numpy 2.1.3, scipy 1.17.1, pandas 3.0.5, statsmodels 0.14.6, matplotlib 3.9.2.

Key artifacts:

- `datasets/phase2_eval_results.csv` — 16-model empirical accuracy data (16-model aggregate evaluation artifacts; validity audit pending/ongoing)
- `datasets/coverage/minimum_valid_population.csv` — outcome-independent selected 22-model roster
- `src/lineage_era/analysis/reml.py` — REML engine (CrossedREML, Woodbury-accelerated)
- `src/lineage_era/analysis/identifiability.py` — Structural and fit-based audit gate
- `src/lineage_era/occupancy.py` — 47-model population definition
- `src/results/phase1/` — D1/D2/D3/liability simulation outputs
- `results/phase2_empirical/audit_summary.json` — Gate diagnostic results

Known invalid artifacts:

- `datasets/eval_samples/*.jsonl` — All 16 files contain non-informative constant predictions (answer 0 for every question). NOT usable for per-question analysis.
- `results/phase2_empirical/similarity_matrix_phi.csv` — All 1.0 (degenerate, from corrupted JSONL).
- `results/phase2_empirical/error_matrix_binary.csv` — All 1s (from corrupted JSONL).

Thresholds: $\kappa_{\max} = 100$, $\mathrm{VIF}_{\max} = 10$, $\mathrm{BLUP}_r = 0.9$, $\mathrm{SE}_{\text{inflation}} = 1.0$.

Seeds: Simulation uses recorded seeds per scenario (A=101, B=202, C=303).

Verification note: All numerical claims in the empirical identifiability results (rank, κ, VIF, occupancy, LOO diagnostics, diagnostic REML estimates) were independently verified by the authors against the source code and stored outputs. The design-space sweep results (105 evaluated configurations; 40 full rank; 3 κ-passing; 3 all-gate passing) and era-resolution half-year/year results were reproduced from the analysis scripts; the full per-configuration output is saved in `results/design_space/sweep_results.csv`.

XIV. CONCLUSION

We formalized the identifiability conditions for crossed family-proxy-era variance decomposition of foundation-model performance—crossed design, full rank, bounded κ, and bounded VIF—and developed a pre-measurement gating procedure that detects rank deficiency and collinearity before any variance components are estimated. The simulation study validated the REML estimator under identifiable and non-identifiable designs and confirmed that three complementary detectors flag nested aliasing in 100% of repetitions.

Applied to 16 real foundation models, the gate fails four of five conditions: the design is rank-deficient (rank 14 of 15), numerically unstable ($\kappa = 4.7 \times 10^{16}$), and has infinite VIF. The failure is structural (Llama alone in 2023Q1). The intended variance decomposition is therefore not identifiable on this population and is not interpreted. A systematic design-space analysis identifies a sufficient design configuration in the evaluated grid (30 models, 6 families, 8 eras). The main result is that the proposed gate detects non-identifiability before inference is attempted.

ACKNOWLEDGMENT

This work was supported in part by the Department of Computer Science and Engineering, SRM Institute of Science and Technology, Tiruchirappalli, India, and in part by the Research Institute of IoT Cybersecurity, National Kaohsiung University of Science and Technology, Kaohsiung, Taiwan.

AI Disclosure: AI language tools were used for code generation and manuscript formatting. All scientific content, experimental design, data analysis, and conclusions were performed and verified by the human authors.

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APPENDIX A DIAGNOSTIC VARIANCE-COMPONENT ESTIMATE

Figure 14 reports diagnostic variance-component estimates obtained by fitting the variance-component model to the non-identifiable 16-model design. The original bootstrap draws were not archived; the reported interval is the historical result as originally recorded. A regenerated bootstrap (panel (b), seed = 42) is shown for reference only and is not the primary result. The estimates are not scientifically interpretable and are provided for completeness only.

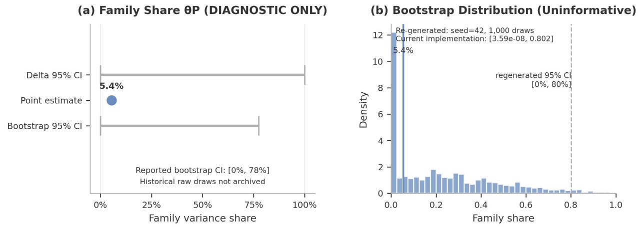

FIGURE 14: Diagnostic variance-component estimates under the invalid (non-identifiable) 16-model design. Panel (a): historical bootstrap interval as originally reported (raw draws not archived). Panel (b): **regenerated sensitivity analysis for reference only — NOT the primary result.** The 95% confidence interval for the family share covers $[0\%, 100\%]$, reflecting the non-identifiability of the rank-deficient design. The near-zero era estimate is an aliasing artifact, not evidence of no era effect. These estimates are not scientifically interpretable.

APPENDIX B OUTCOME-INDEPENDENT CANDIDATE POPULATION

TABLE XIII lists the 22 models selected by the outcome-independent procedure from the 47-model candidate population. The selection satisfies all structural gate conditions (S1–S3, N2) within the candidate frame, but does not satisfy the condition-number gate ($\kappa = 1100 > 100$) and is therefore not a fully gate-passing population.

TABLE XIII: Outcome-Independent Candidate Population (22 Models). Candidate subset from the 47-model frame, selected by the outcome-independent procedure. All 6 families are represented with ≥ 2 models each. The selection crossing constraint requires every family to span ≥ 2 eras and at least 2 eras to contain ≥ 2 families; this subset satisfies both. This subset satisfies structural conditions (S1–S3, N2) but not the condition-number gate ($\kappa = 1100 > 100$).

Model	Family	Era	Params	Access	Measured?
Llama-1	Llama	2023Q1	7B	Public	Yes
Llama-3.1-8B	Llama	2024Q3	8B	Public	Yes
Llama-3.3-70B	Llama	2024Q4	70B	Public	No
Qwen-7B	Qwen	2023Q3	7B	Public	Yes
Qwen-1.5-72B	Qwen	2024Q1	72B	Public	No
Mistral-7B	Mistral	2023Q3	7B	Public	Yes
Mistral-Small-3	Mistral	2025Q1	24B	Public	Yes
Mistral-Small-3.1	Mistral	2025Q1	24B	Public	Yes
Mistral-Small-3.2	Mistral	2025Q2	24B	Public	Yes
Devstral-2	Mistral	2025Q4	24B	Public	Yes
Mistral-Small-4	Mistral	2026Q1	119B	Gated	Yes
Phi-1	Phi	2023Q2	1.3B	Public	Yes
Phi-1.5	Phi	2023Q3	1.3B	Public	Yes
Phi-2	Phi	2023Q4	2.7B	Public	Yes
Phi-3	Phi	2024Q2	14B	Public	Yes
Phi-4	Phi	2024Q4	14B	Public	Yes
Phi-4-reas.+	Phi	2025Q2	14B	Public	Yes
Phi-4-reas.-v.	Phi	2026Q1	15B	Public	No
Gemma-3n	Gemma	2025Q2	4B	Public	Yes
Gemma-4-12B	Gemma	2026Q2	12B	Public	Yes
DeepSeek-V3.1	DeepSeek	2025Q3	685B	Public	No
DeepSeek-V3.2	DeepSeek	2025Q4	685B	Public	No

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